

Notes on Linear Regression

Luis Mendez

I. ASSUMPTIONS OF LINEAR REGRESSION

- Linearity: The relationship between the independent and dependent variable is linear.
- Independence: The residuals (errors) are independent of each other.
- Homoscedasticity: The residuals have constant variance at every level of the independent variable.
- Normality: The residuals are normally distributed.
- Multivariate Normality: The independent variables are normally distributed (for multiple regression).
- No Multicollinearity: The independent variables are not highly correlated with each other.

